

VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY

Ho Chi Minh City University of Technology

Faculty of Computer Science and Engineering



COMPUTER NETWORKS - LAB (CO3094)

Laboratory

Report Lab 3

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Semester: 251

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HO CHI MINH CITY, NOVEMBER 2025



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1 Wireshark Lab: ICMP

1.1 ICMP and Ping

Question 1. What is the IP address of your host? What is the IP address of the destination host?
The IP address of my host is 192.168.1.177. The IP address of the destination host is 143.89.209.9.

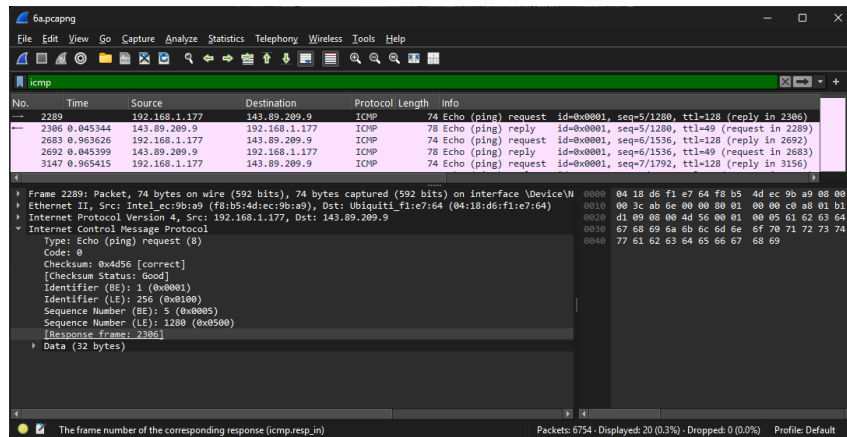


Figure 1: Echo request packet showing the source IP address 192.168.1.177 and destination IP address 143.89.209.9. The ICMP header details reveal this is an Echo (ping) request message.

Question 2. Why is it that an ICMP packet does not have source and destination port numbers?

ICMP is a network layer protocol that operates on top of IP. Since it is not a transport layer protocol, the concept of ports does not exist. Port numbers are only relevant for transport layer protocols such as TCP and UDP, which multiplex data from different applications.

Question 3. Examine one of the ping request packets sent by your host. What are the ICMP type and code numbers? What other fields does this ICMP packet have? How many bytes are the checksum, sequence number and identifier fields?

From Figure 1, the ICMP packet captured is an Echo (ping) request, which has a Type value of 8 and a Code value of 0. In addition to these fields, the ICMP Echo Request packet includes a Checksum, an Identifier, and a Sequence Number, each occupying 2 bytes. These fields allow the sender to match requests with replies and detect corruption in transit.

Question 4. Examine the corresponding ping reply packet. What are the ICMP type and code numbers? What other fields does this ICMP packet have? How many bytes are the checksum, sequence number and identifier fields?

The corresponding ICMP packet is an Echo (ping) reply, which has a Type value of 0 and a Code value of 0. Similar to the request packet, it contains the Checksum, Identifier, and Sequence Number fields, each of which is 2 bytes in length. The Identifier and Sequence Number in the reply match those in the request, allowing the ping utility to correlate responses with the original requests.

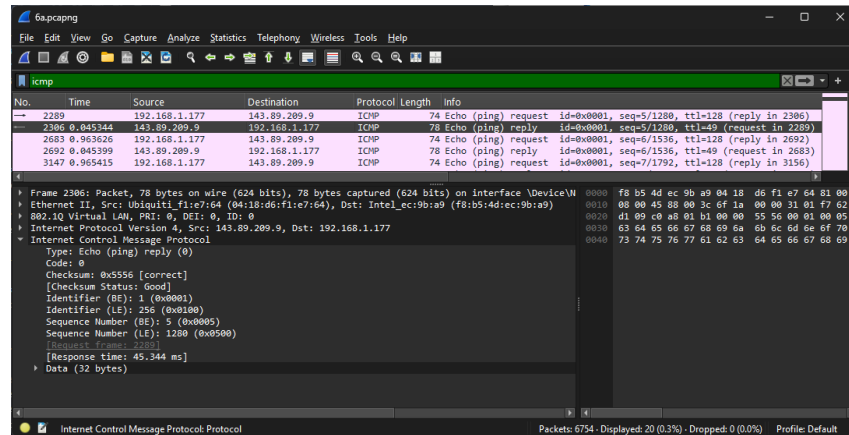


Figure 2: Echo reply packet with ICMP Type 0 (Echo Reply) and Code 0. The Identifier and Sequence Number match the corresponding request, enabling proper correlation of ping responses.

1.2 ICMP and Traceroute

Question 5. What is the IP address of your host? What is the IP address of the target destination host?

The IP address of my host is 192.168.1.10. The IP address of the destination host is 143.89.209.9.

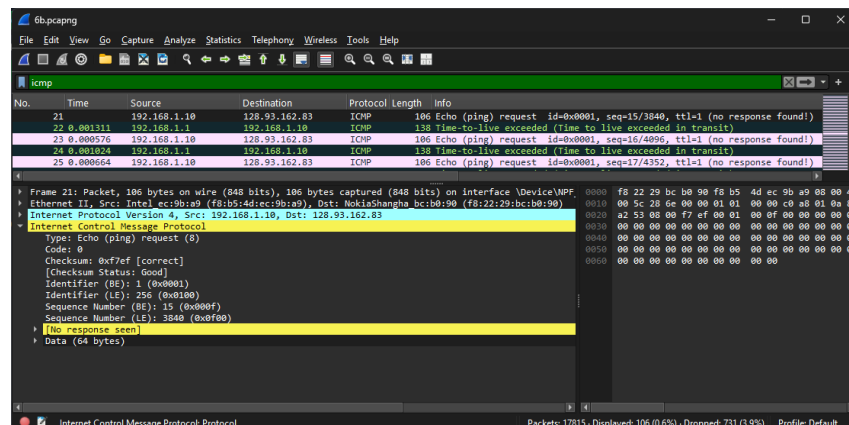


Figure 3: Echo request packet from traceroute showing source IP 192.168.1.10 and destination 143.89.209.9. The TTL field is gradually incremented in successive probes to discover intermediate routers.

Question 6. If ICMP sent UDP packets instead (as in Unix/Linux), would the IP protocol number still be 01 for the probe packets? If not, what would it be?

If ICMP used UDP packets instead, the IP protocol number would not be 01 (which is for ICMP). It would be 17, corresponding to UDP. The protocol number in the IP header identifies which upper-layer protocol should process the payload.

Question 7. Examine the ICMP echo packet in your screenshot. Is this different from the ICMP ping query packets in the first half of this lab? If yes, how so?

From Figure 3, the packets themselves are not particularly different in structure, but the packets in the traceroute gradually increase their TTL value. Traceroute starts with TTL=1 for the first probe, TTL=2 for the second, and so on, causing each packet to expire at successive routers along the path, which then return ICMP Time Exceeded messages.

Question 8. Examine the ICMP error packet in your screenshot. It has more fields than the ICMP echo packet. What is included in those fields?

We can see that the ICMP error packet (Type 11 – Time Exceeded) contains an extra field "Unused: 00000000", followed by the IP header of the original request (Src: 192.168.1.101, Dst: 138.96.146.2) and the first 8 bytes of the ICMP Echo Request (Type 8). This encapsulation allows the sender to identify which original packet triggered the error message.

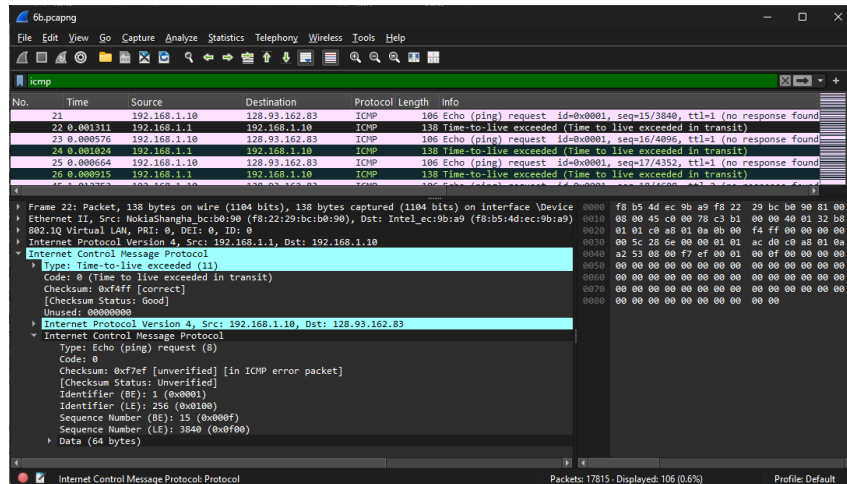


Figure 4: ICMP Time Exceeded error packet (Type 11) containing the IP header and first 8 bytes of the original datagram that triggered the error. This information helps the sender identify which probe caused this response.

Question 9. Examine the last three ICMP packets received by the source host. How are these packets different from the ICMP error packets? Why are they different?

Since the Request packet has reached its destination, a reply packet is returned, not an error. The destination host responds with ICMP Echo Reply (Type 0) messages instead of Time Exceeded errors, indicating successful delivery to the final destination.

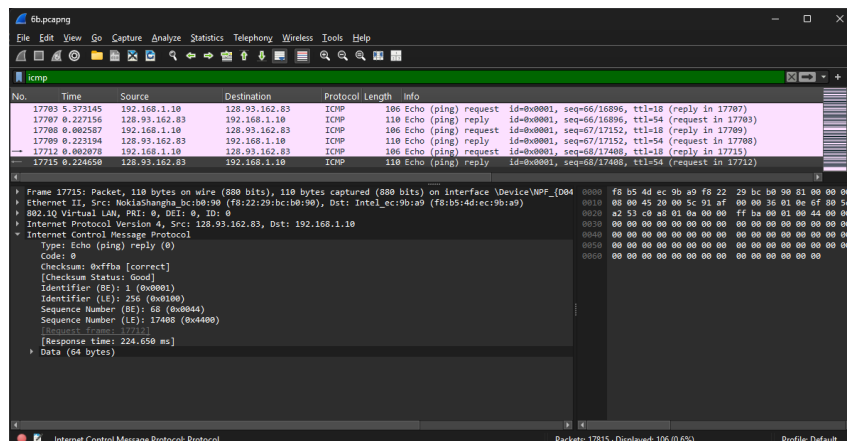


Figure 5: Echo reply packet (Type 0) from the final destination host. Unlike intermediate routers that send Time Exceeded messages, the destination responds with Echo Replies when traceroute probes reach their target.

Question 10. Within the traceroute measurements, is there a link whose delay is significantly longer than others? On the basis of the router names, can you guess the location of the two routers on the end of this link?

The traceroute shows a significant RTT jump between hop 9 (att/NTT node in Hong Kong) and hop 10 (vienat02 — a Vienna/European backbone node). Based on the router names, this link appears to be an intercontinental link connecting Hong Kong (Asia) to Vienna / central Europe, which explains the large increase in delay due to the physical distance and light propagation time.

```

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Windows\System32> tracert www.inria.fr

Tracing route to inria.fr [128.93.162.83]
over a maximum of 30 hops:

  0  1 ms  1 ms  1 ms  192.168.0.1
  1  9 ms  4 ms  7 ms  118.69.185.49
  2  6 ms  3 ms  *  118.69.253.233
  3  4 ms  4 ms  3 ms  42.112.0.76
  4  5 ms  6 ms  4 ms  42.112.0.77
  5  *  24 ms  22 ms  42.112.0.192
  6  44 ms  43 ms  42 ms  118.69.253.73
  7  46 ms  48 ms  48 ms  ae-7.a01.chwahk03.hk.bb.gin.ntt.net [157.238.227.36]
  8  45 ms  43 ms  43 ms  ae-14.r28.tkokh01.hk.bb.gin.ntt.net [129.250.2.188]
  9  291 ms  304 ms  305 ms  ae-6.r22.vienat02.at.bb.gin.ntt.net [129.250.4.81]
 10  213 ms  244 ms  212 ms  ae-2.a01.vienat02.at.bb.gin.ntt.net [129.250.7.39]
 11  *  316 ms  *  ael3.cr3-vie2.ip4.gtt.net [212.222.122.89]
 12  *  *  291 ms  ae7.cr6-par11.ip4.gtt.net [213.200.120.85]
 13  *  *  317 ms  ip4.gtt.net [212.222.6.69]
 14  324 ms  *  334 ms  hu0-4-0-ren-nr-orsay-rtr-091.noc.renater.fr [193.51.180.131]
 15  *  290 ms  374 ms  inria-rocquencourt-vl1631-tel-4-inria-rtr-021.noc.renater.fr [193.51.184.177]
 16  *  296 ms  293 ms  unit240-reth1-vfw-ext-dcl.inria.fr [192.93.122.19]
 17  368 ms  305 ms  293 ms  prod-inriafr-cms.inria.fr [128.93.162.83]

Trace complete.
PS C:\Windows\System32>

```

Figure 6: Traceroute measurement results showing round-trip times for each hop. A significant delay increase is visible at the intercontinental link between Asian and European network segments.

2 Wireshark Lab: Ethernet and ARP

2.1 Capturing and analyzing Ethernet frames

Question 1. What is the 48-bit Ethernet address of your computer?

The 48-bit Ethernet address of my computer is BelkinIntern_75:b1:52 (c4:41:1e:75:b1:52). This MAC address uniquely identifies the network interface card in my computer on the local Ethernet segment.

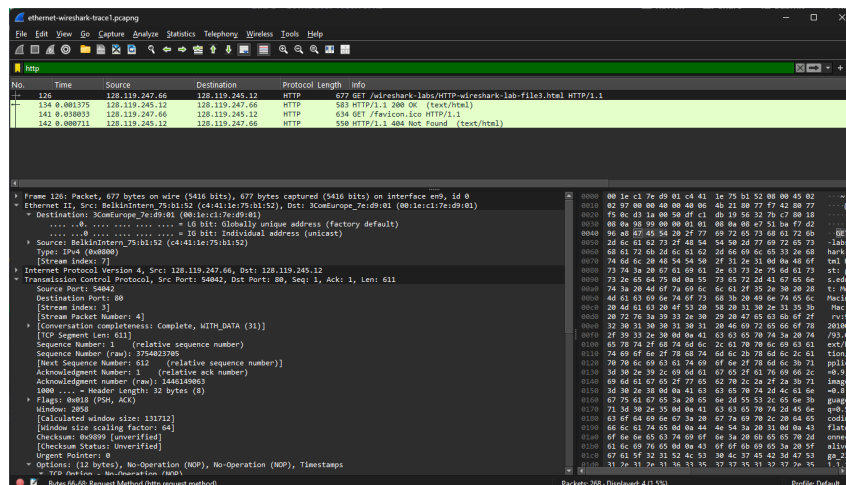


Figure 7: Ethernet frame containing HTTP request. The source MAC address c4:41:1e:75:b1:52 identifies the sender's network interface, while the frame type field indicates IPv4 payload.

Question 2. What is the 48-bit destination address in the Ethernet frame? Is this the Ethernet address of gaia.cs.umass.edu? (Hint: the answer is no). What device has this as its Ethernet address?

From Figure 7, the 48-bit destination address in the Ethernet frame is 3ComEurope_7e:d9:01 (00:1e:c1:7e:d9:01). This is not the Ethernet address of gaia.cs.umass.edu. Instead, it is the MAC address of the first-hop router (default gateway) on the local network. Ethernet operates at the link layer and can only deliver frames to devices on the same local network segment.

Question 3. What is the hexadecimal value for the two-byte Frame type field in the Ethernet frame carrying the HTTP GET request? What upper layer protocol does this correspond to?

From Figure 7, the hexadecimal value for the two-bytes Frame type field is IPv4 (0x0800). This Ethernet-Type value indicates that the payload of this Ethernet frame is an IPv4 datagram.

Question 4. How many bytes from the very start of the Ethernet frame does the ASCII "G" in "GET"

appear in the Ethernet frame? Do not count any preamble bits in your count, i.e., assume that the Ethernet frame begins with the Ethernet frame's destination address.

From Figure 7, there are **66 bytes**. This includes the 14-byte Ethernet header (6 bytes destination MAC + 6 bytes source MAC + 2 bytes type), 20-byte IP header, 20-byte TCP header, and 12 bytes of preceding HTTP header fields before the "G" character.

Question 5. What is the value of the Ethernet source address? Is this the address of your computer, or of gaia.cs.umass.edu (Hint: the answer is no). What device has this as its Ethernet address?

The value of the Ethernet source address is **3ComEurope_7e:d9:01 (00:1e:c1:7e:d9:01)**. It is the address of the first router (default gateway). When the HTTP response arrives at the local network, the router replaces its WAN-side source MAC with its LAN-side MAC before forwarding to the destination host.

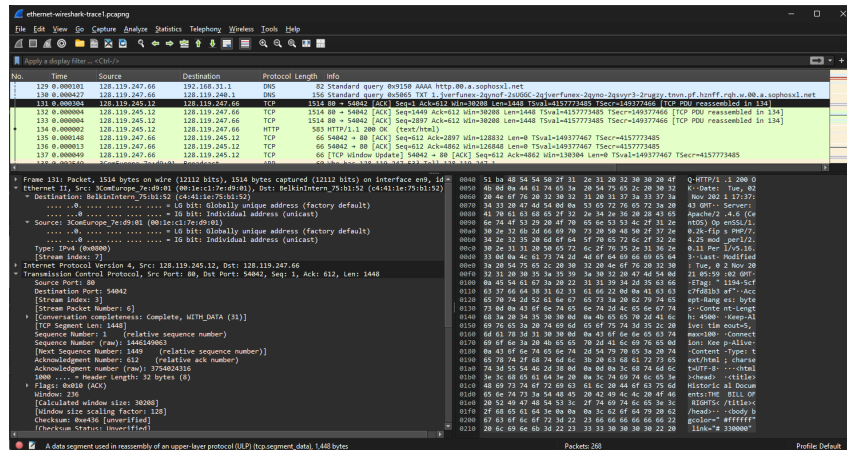


Figure 8: Ethernet frame containing the first byte of HTTP response. The source MAC address belongs to the router, which acts as an intermediary forwarding traffic between the local network and the Internet.

Question 6. What is the destination address in the Ethernet frame? Is this the Ethernet address of your computer?

From Figure 8, the destination address in the Ethernet frame is **BelkinIntern_75:b1:52 (c4:41:1e:75:b1:52)** which is the address of my computer. The router correctly forwards the response to the original requester's MAC address.

Question 7. Give the hexadecimal value for the two-byte Frame type field. What upper layer protocol does this correspond to?

From Figure 8, the hexadecimal value of the two bytes frame type field is **IPv4 (0x0800)**, indicating the payload is an IPv4 packet.

Question 8. How many bytes from the very start of the Ethernet frame does the ASCII "O" in "OK" (i.e., the HTTP response code) appear in the Ethernet frame? Do not count any preamble bits in your count, i.e., assume that the Ethernet frame begins with the Ethernet frame's destination address.

From Figure 8, there are **79 bytes**. This offset accounts for the Ethernet header, IP header, TCP header, and the preceding characters in the HTTP response status line before "OK".

Question 9. How many Ethernet frames (each containing an IP datagram, each containing a TCP segment) carry data that is part of the complete HTTP "OK 200 ..." reply message?

From Figure 8, there are **4 Ethernet frames** (131, 132, 133, 134) that carry data as part of the complete HTTP "OK 200 ..." reply message. The HTTP response is segmented across multiple TCP segments due to Maximum Segment Size (MSS) limitations, with each segment encapsulated in its own IP datagram and Ethernet frame.

2.2 The Address Resolution Protocol

Question 10. How many entries are stored in your ARP cache?

My ARP cache stores **15 entries** in total. The ARP cache maintains a mapping between IP addresses and MAC addresses for recently communicated hosts on the local network to avoid repeated ARP queries.


```

C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.7019]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ACER>arp -a

Interface: 192.168.56.1 --- 0x5
Internet Address      Physical Address      Type
192.168.56.255        ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static

Interface: 192.168.1.10 --- 0x11
Internet Address      Physical Address      Type
192.168.1.1           f8-22-29-bc-b0-90    dynamic
192.168.1.2           2a-00-0e-c5-62-d3    dynamic
192.168.1.8           b0-1e-a0-c5-c7-37    dynamic
192.168.1.13          ea-45-af-c3-8d-8a    dynamic
192.168.1.255         ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static

C:\Users\ACER>

```

Figure 9: Output of the `arp -a` command displaying all entries in the ARP cache. Each entry maps an IP address to its corresponding physical (MAC) address.

Question 11. What is contained in each displayed entry of the ARP cache?

From Figure 9, each entry contains the IP address, MAC address, and type of entry (dynamic or static). Dynamic entries are learned through ARP protocol exchanges, while static entries are manually configured.

Question 12. What is the hexadecimal value of the source address in the Ethernet frame containing the ARP request message sent out by your computer?

The hexadecimal value of the source address in the Ethernet frame containing the ARP request message sent out by your computer is BelkinIntern_75:b1:52 (c4:41:1e:75:b1:52). This identifies which host is requesting the MAC address resolution.

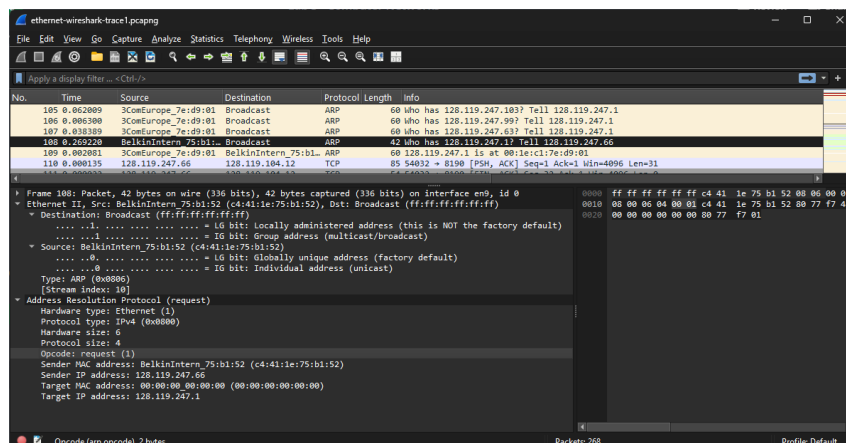


Figure 10: ARP request message broadcast to all hosts on the subnet. The source MAC identifies the requester, while the destination is set to the broadcast address to reach all devices.

Question 13. What is the hexadecimal value of the destination addresses in the Ethernet frame containing the ARP request message sent out by your computer? And what device (if any) corresponds to that address (e.g., client, server, router, switch or otherwise...)?

From Figure 10, the hexadecimal value of the destination address in the Ethernet frame containing the ARP request message sent out by your computer is Broadcast (ff:ff:ff:ff:ff:ff) (All hosts in that subnet). ARP requests must be broadcast because the sender does not yet know the target's MAC address.

Question 14. What is the hexadecimal value for the two-byte Ethernet Frame type field. What upper layer protocol does this correspond to?

From Figure 10, the hexadecimal value of the two-byte Ethernet Frame type field is ARP (0x0806). This EtherType indicates the frame carries an ARP packet rather than an IP datagram.

Question 15. How many bytes from the very beginning of the Ethernet frame does the ARP opcode field begin?

From Figure 10, there are **20 bytes**. This offset includes the 14-byte Ethernet header plus the first 6 bytes of the ARP header (hardware type, protocol type, and hardware/protocol address lengths).

Question 16. What is the value of the opcode field within the ARP request message sent by your computer?

From Figure 10, the value of the opcode field within the ARP request message is **request (1)**. An opcode of 1 indicates this is an ARP request seeking a MAC address, while opcode 2 would indicate an ARP reply.

Question 17. Does the ARP request message contain the IP address of the sender? If the answer is yes, what is that value?

From Figure 10, yes and sender IP address is **128.119.247.66**. The sender's IP and MAC addresses are included in the ARP request to allow the target to update its ARP cache and send a direct reply.

Question 18. What is the IP address of the device whose corresponding Ethernet address is being requested in the ARP request message sent by your computer?

From Figure 10, the target IP address is **128.119.247.1**. This is typically the default gateway or another host on the local subnet that the sender needs to communicate with.

Question 19. What is the value of the opcode field within the ARP reply message received by your computer?

The value of the opcode field within the ARP reply message received is **reply (2)**. This indicates the message is an ARP response providing the requested MAC address.

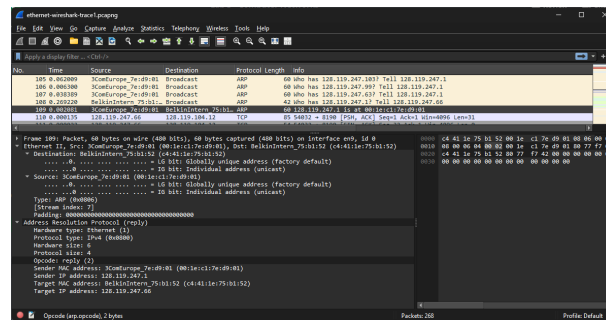


Figure 11: ARP reply message with opcode 2, providing the MAC address corresponding to the requested IP address. The reply is sent directly to the original requester, not broadcast.

Question 20. Finally(!), let's look at the answer to the ARP request message! What is the Ethernet address corresponding to the IP address that was specified in the ARP request message sent by your computer (see question 18)?

From Figure 11, the sender MAC address is **3ComEurope_7e:d9:01 (00:1e:c1:7e:d9:01)**. This MAC address is now cached by the requesting host, allowing future communications to this IP address without additional ARP queries until the cache entry expires.

Question 21. We've looked the ARP request message sent by your computer running Wireshark, and the ARP reply message sent in response. But there are other devices in this network that are also sending ARP request messages that you can find in the trace. Why are there no ARP replies in your trace that are sent in response to these other ARP request messages?

This is because the request message is broadcast, but the reply message is only sent to the requested host. When other devices send ARP requests, the replies are unicast directly to those requesters' MAC addresses, so they are not visible to other hosts on the network (unless the network interface is in promiscuous mode and captures all traffic).

3 Wireshark Lab: 802.11 WiFi

3.1 Beacon Frames

Question 1. What are the SSIDs of the two access points that are issuing most of the beacon frames in this trace?

The SSIDs of the two access points that are issuing most of the beacon frames are 30 Munroe St and linksys12t. These SSIDs identify the wireless networks being advertised by nearby access points.

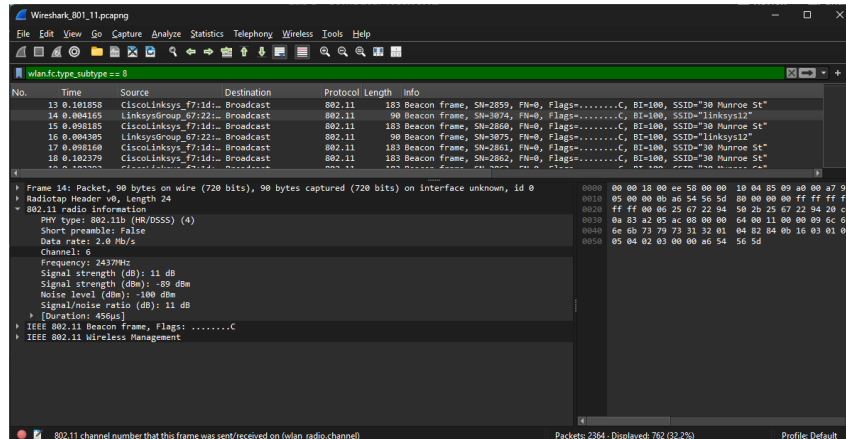


Figure 12: 802.11 radio information from beacon frame showing channel and other physical layer parameters. Access points periodically broadcast beacons to announce their presence and capabilities.

Question 2. What 802.11 channel is being used by both of these access points?

From Figure 12, channel 6 is being used by both of these access points. In the 2.4 GHz band, channels 1, 6, and 11 are commonly used as they are the only non-overlapping channels in this frequency range.

Question 3. What is the interval of time between the transmissions of beacon frames from this access point (AP)?

The interval of time between the transmissions of beacon frames from this access point (AP) is 0.102400 seconds (approximately 102.4 milliseconds). This is the standard beacon interval, known as a Time Unit (TU), with 1 TU = 1024 microseconds.

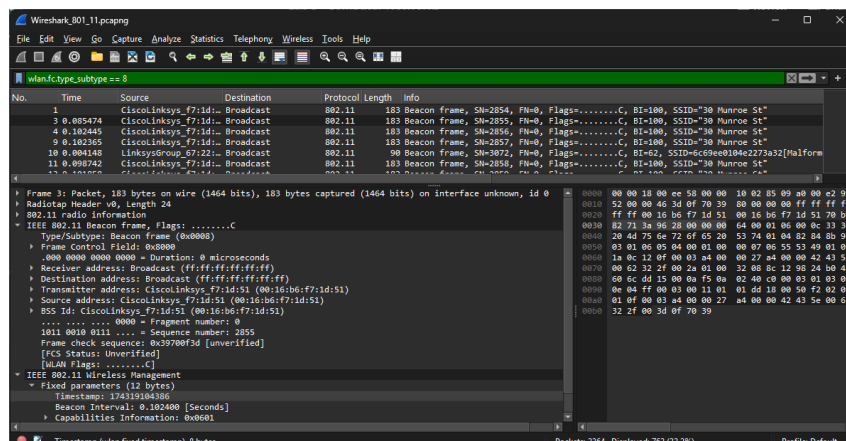


Figure 13: 802.11 Beacon frame containing network parameters including SSID, supported rates, and security capabilities. The beacon interval field specifies the time between successive beacon transmissions.

Question 4. What (in hexadecimal notation) is the source MAC address on the beacon frame from this access point?

From Figure 13, the source MAC address is CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51). This is the MAC address of the access point's radio interface that is transmitting the beacon.

Question 5. What (in hexadecimal notation) is the destination MAC address on the beacon frame from

30 Munroe St?

From Figure 13, the destination MAC address is **Broadcast** (ff:ff:ff:ff:ff:ff). Beacon frames are always broadcast so that all wireless devices in range can discover the network.

Question 6. What (in hexadecimal notation) is the MAC BSS ID on the beacon frame from 30 Munroe St?

From Figure 13, the MAC BSS ID on the beacon frame from 30 Munroe St is **CiscoLinksys_f7:1d:51** (00:16:b6:f7:1d:51). The BSSID (Basic Service Set Identifier) uniquely identifies this particular BSS and is typically the MAC address of the access point.

Question 7. The beacon frames from the 30 Munroe St access point advertise that the access point can support four data rates and eight additional "extended supported rates." What are these rates?

They are:

- Tag: Supported Rates 1(B), 2(B), 5.5(B), 11(B) [Mbit/sec]
- Tag: Extended Supported Rates 6(B), 9, 12(B), 18, 24(B), 36, 48, 54 [Mbit/sec]

The "(B)" notation indicates basic rates that all stations must support to join this BSS.

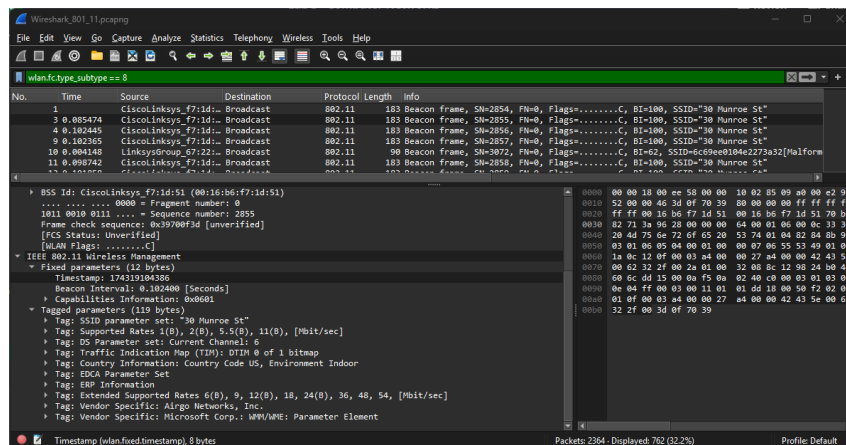


Figure 14: 802.11 Beacon frame details showing both *Supported Rates* and *Extended Supported Rates* tags. These specify the data transmission speeds available in this wireless network.

3.2 Data Transfer

Question 8. Find the 802.11 frame containing the SYN TCP segment for this first TCP session (that downloads *alice.txt*) at *t*=24.8110. What are three MAC address fields in the 802.11 frame? Which MAC address in this frame corresponds to the wireless host (give the hexadecimal representation of the MAC address for the host)? To the access point? To the first-hop router? What is the IP address of the wireless host sending this TCP segment? What is the destination IP address for the TCP syn segment?

- MAC Address:
 - Source address: **Intel_d1:b6:4f** (00:13:02:d1:b6:4f) (Host)
 - Destination address: **CiscoLinksys_f4:eb:a8** (00:16:b6:f4:eb:a8) (Router)
 - BSS Id: **CiscoLinksys_f7:1d:51** (00:16:b6:f7:1d:51) (AP)
- IP address:
 - Source Address: **192.168.1.109**
 - Destination Address: **128.119.245.12**

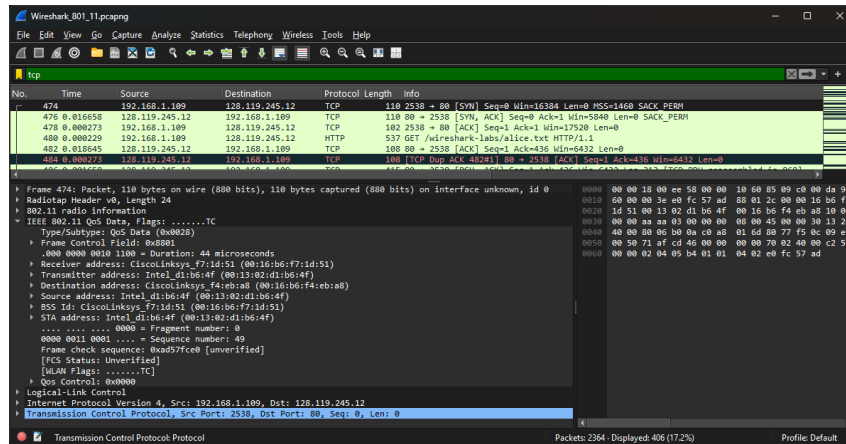


Figure 15: TCP SYN segment encapsulated in 802.11 frame. The three MAC address fields serve different purposes: identifying the wireless sender, the immediate recipient, and the BSS to which they belong.

Question 9. Does the destination IP address of this TCP SYN correspond to the host, access point, first-hop router, or the destination web server?

Yes, it corresponds to the destination web server. The IP addresses identify the ultimate source and destination, while the MAC addresses in the 802.11 header identify devices on the local wireless segment.

Question 10. Find the 802.11 frame containing the SYNACK segment for this TCP session received at $t = 24.8277$. What are three MAC address fields in the 802.11 frame? Which MAC address in this frame corresponds to the host? To the access point? To the first-hop router? Does the sender MAC address in the frame correspond to the IP address of the device that sent the TCP segment encapsulated within this datagram? (Hint: review Figure 6.19 in the text if you are unsure of how to answer this question, or the corresponding part of the previous question. It's particularly important that you understand this).

- Source address: CiscoLinksys_f4:eb:a8 (00:16:b6:f4:eb:a8) (Router)
- Destination address: 91:2a:b0:49:b6:4f (91:2a:b0:49:b6:4f) (Host)
- BSS Id: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51) (AP)
- A Sender's MAC address corresponds to a router, while an IP address corresponds to a web server.

The MAC address identifies the device on the local wireless segment (the router), while the IP address in the encapsulated packet identifies the actual source of the TCP segment (the web server). This demonstrates the difference between link-layer and network-layer addressing.

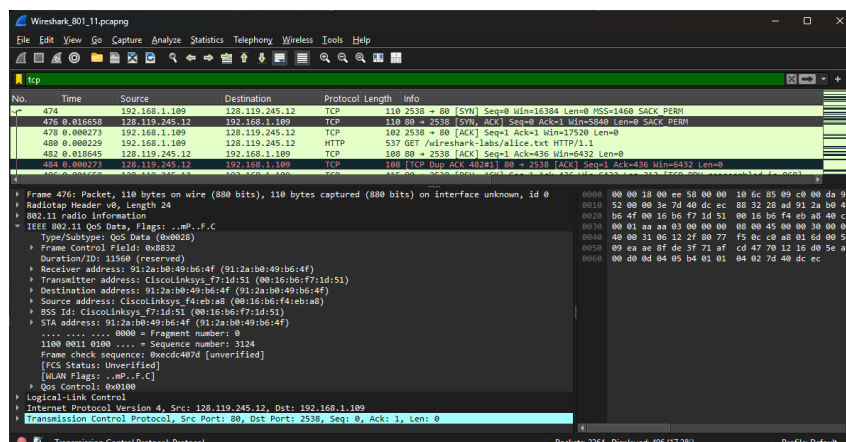


Figure 16: TCP SYN-ACK segment in 802.11 frame. Note that the 802.11 source MAC is the router, but the IP source address inside is the web server, illustrating the multi-hop nature of Internet communication.

3.3 Disassociation/Authentication/Association

Question 11. What two actions are taken (i.e., frames are sent) by the host in the trace just after $t = 49$, to end the association with the 30 Munroe St AP that was initially in place when trace collection began? (Hint: one is an IP-layer action, and one is an 802.11-layer action).

Send DHCP Release and send Deauthentication are taken by the host in the trace just after $t = 49$, to end the association with the 30 Munroe St AP. The DHCP Release returns the IP address to the DHCP server, while the Deauthentication frame terminates the 802.11 association with the access point.

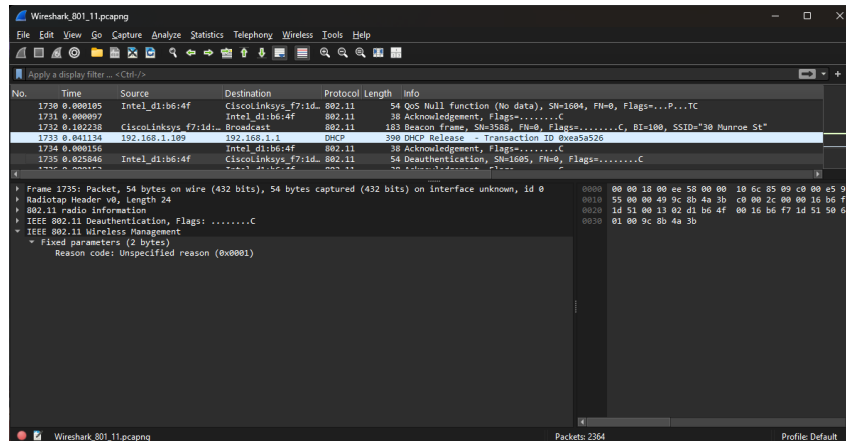


Figure 17: Frames showing the host ending its association with the AP. The DHCP Release operates at the network layer, while Deauthentication is an 802.11 management frame at the link layer.

Question 12. Let's look first at AUTHENTICATION frames. At $t = 63.1680$, our host tries to associate with the 30 Munroe St AP. Use the Wireshark display filter `wlan.fc.subtype == 11` to show AUTHENTICATION frames sent from the host to and AP and vice versa. What form of authentication is the host requesting?

The host requesting Open System (0) in Authentication Algorithm. Open System authentication is the simplest 802.11 authentication method where any device can authenticate without providing credentials.

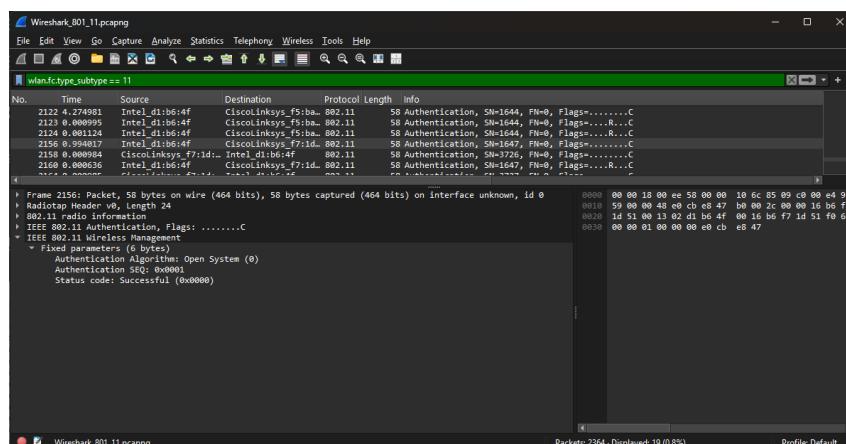


Figure 18: Authentication request frame with Open System authentication algorithm. This is the first step in establishing an 802.11 connection before association can occur.

Question 13. What is the Authentication SEQ value (authentication sequence number) of this authentication frame from host to AP?

From Figure 18, the Authentication SEQ value is 0x0001. The authentication transaction uses a two-step sequence: the request has sequence number 1, and the response has sequence number 2.

Question 14. The AP response to the authentication request is received at $t = 63.1690$. Has the AP accepted the form of authentication requested by the host?

Yes because the value of Status code is **Successful** (0x0000). A status code of 0x0000 indicates the authentication was accepted, allowing the host to proceed to the association phase.

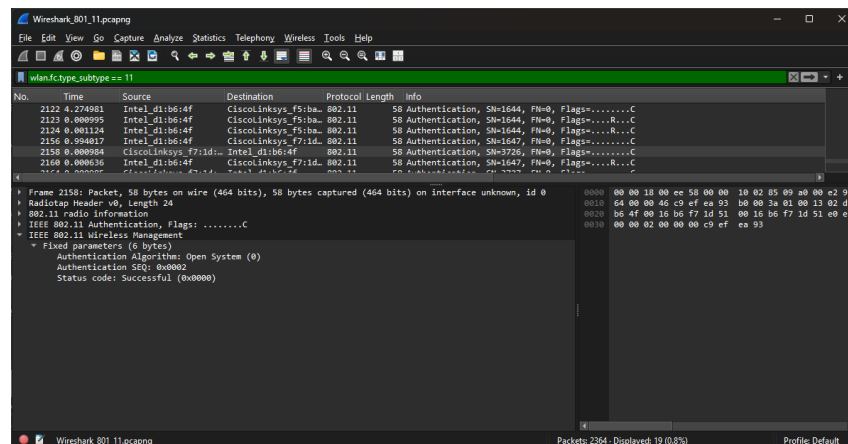


Figure 19: Authentication response frame from AP showing successful status code. The AP has accepted the host's authentication request using Open System authentication.

Question 15. What is the Authentication SEQ value of this authentication frame from AP to Host? From Figure 19, the Authentication SEQ value of this authentication frame from AP to Host is 0x0002. This completes the two-way authentication handshake.

Question 16. What rates are indicated in the frame as SUPPORTED RATES. Do not include in your answers below any rates that are indicates as EXTENDED SUPPORTED RATES. The Supported rates are 1(B), 2(B), 5.5(B), 11(B), 6(B), 9, 12(B), 18 [Mbit/sec]. These are the data rates the host can support for communication with this BSS.

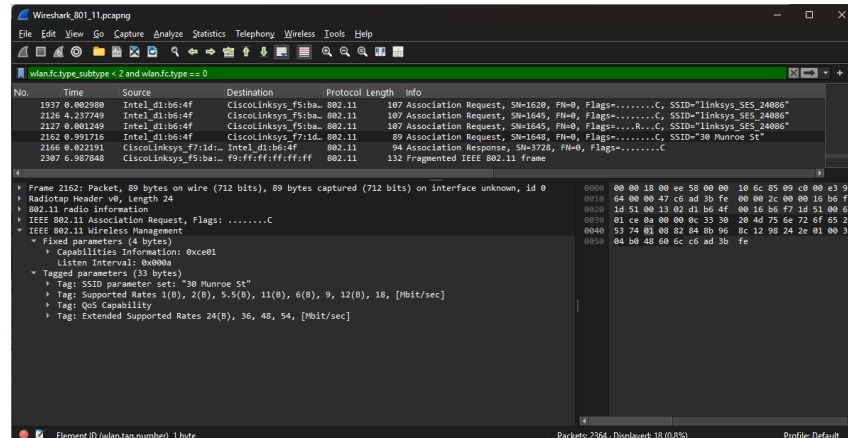


Figure 20: Association request frame containing the host's supported data rates and other capabilities. The AP uses this information to determine compatible communication parameters.

Question 17. Does the ASSOCIATION RESPONSE indicate a Successful or Unsuccessful association response?

Yes because the value of status code is **Successful** (0x0000). The host is now fully associated with the AP and can begin exchanging data frames.

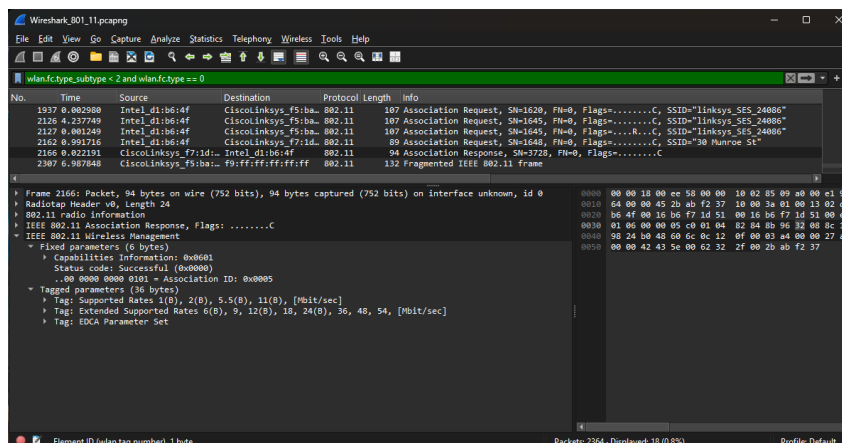


Figure 21: Association response frame from the AP with successful status code. The host has completed the authentication and association process and can now communicate through this AP.

Question 18. Does the fastest (largest) Extended Supported Rate the host has offered match the fastest (largest) Extended Supported Rate the AP is able to provide?

From Figure 21, yes and it is 54 Mbit/s. This is the maximum data rate for 802.11g networks, which both the host and AP support.

Code Availability

The complete implementation of this lab session, including all source code and related files, is available on GitHub. Please access the repository via <https://github.com/longlephamtien/CN251-Lab>.